

NGUYEN TUNG

[GitHub](#) ◇ [LinkedIn](#)

AI Engineer/Researcher

Hanoi, Vietnam ◇ (+84) 969 200 308 ◇ [Email](#)

EDUCATION

Hanoi University of Science and Technology Master of Science in Applied Mathematics and Informatics Graduation Classification: Excellent Cumulative GPA: 3.81/4.0	09/2023 – 12/2025
Hanoi University of Science and Technology Engineer Degree in Applied Mathematics and Informatics Graduation Classification: Very Good Cumulative GPA: 3.46/4.0	09/2017 – 05/2022
Chu Van An National High School High School Diploma	09/2014 – 05/2017

SKILLS

Research Areas	Computer Vision, Intelligent Document Processing, Generative AI, Reinforcement Learning.
Programming	Python, C#, C/C++, MATLAB.
AI/ML Frameworks	PyTorch, TensorFlow, MLflow, vLLM.
Computer Vision	Object Detection, OCR, Image Classification, Tracking, Segmentation, Layout Analysis.
Generative AI	Large Language Models, Vision-Language Models, RAG, Prompt Engineering, Model Fine-tuning.
MLOps & Tools	Docker, Git, Triton Inference Server, Label Studio, Roboflow, Dify, Ollama, N8N.
Professional Skills	Research, Technical Presentation, Teamwork, Leadership, Scrum.

WORK EXPERIENCE

AI Engineer, Viettel Solutionxns Corp., Full-time	03/2022 – Present
- Research and develop Intelligent Document Processing (IDP) systems for extracting structured information from complex business documents, including forms, tables, scanned documents, and semi-structured layouts. - Fine-tune vision-language models (VLMs) for document information extraction tasks, improving the ability to understand document layouts, recognize key fields, and generate structured outputs from visual inputs. - Develop layout analysis and table extraction modules to detect document regions, preserve reading order, and support downstream OCR, information extraction, and validation workflows. - Build end-to-end data processing and inference pipelines for license plate recognition, covering image preprocessing, vehicle and plate detection, OCR, post-processing, and result validation. - Deploy license plate recognition models with Triton Inference Server and optimize inference workflows for reliable integration into traffic monitoring and violation detection systems. - Research and develop traffic violation behavior recognition features by combining detection, tracking, license plate recognition, and rule-based video analytics. - Research and develop speech enhancement models to improve audio quality and user experience for the Viettel Meeting Platform.	
Research Consultant, WorldQuant BRAIN, Part-time	03/2025 – Present
Research, design, and simulate quantitative financial alpha signals on the WorldQuant BRAIN platform using data-driven investment hypotheses.	

- Researched and evaluated face processing models, including face detection, face alignment, and face recognition components for real-time computer vision applications.
- Studied object tracking algorithms for camera-based monitoring systems and analyzed tracking performance under challenging conditions such as occlusion, motion blur, and crowded scenes.
- Supported the development of a camera monitoring system by preparing datasets, running model experiments, and integrating computer vision modules into the application pipeline.
- Conducted performance evaluation and error analysis to improve model reliability, inference speed, and robustness in practical surveillance scenarios.

- Researched facial expression recognition models and multitask learning approaches for jointly learning facial attributes from image data.
- Built and supported experiments for face recognition tasks, including dataset preparation, model training, evaluation, and comparison of recognition performance.
- Annotated, cleaned, and organized image datasets to improve data quality for training and validation workflows.
- Visualized image datasets, label distributions, and model prediction results to support error analysis and guide model improvement.

- Researched and implemented CenterNet-based object detection models to localize Vietnamese ID cards in eKYC images.
- Developed card alignment workflows using detected key regions to crop, rotate, and normalize ID card images before text recognition.
- Applied OCR tools to recognize and extract textual information from Vietnamese ID cards, supporting identity verification use cases.
- Tested the detection, alignment, and OCR pipeline on ID card images captured under different lighting, angle, and background conditions.

AWARDS

Tuổi trẻ sáng tạo toàn quốc awarded by Ho Chi Minh Communist Youth Union 2025.

Academic Scholarship awarded by business partners of HUST 2020, 2021.

PUBLICATIONS

VPHQA: Vietnamese Pregnancy Health Question Answering dataset, EAI INISCOM 2025. DOI 10.1007/978-3-032-02362-9_13.

Chatbot and Reproductive Health for Minority Women, AI4SDF 2025.

The Solution of Computer Vision for Combating Covid-19, Computational Intelligence Techniques for Green Smart Cities, Springer, 2021. DOI: 10.1007/978-3-030-96429-0_9.

PERSONAL PROJECTS

dMOM Chatbot

- Developed a medical chatbot for the dMOM Project, sponsored by PHAD under the Ministry of Health, to support reproductive and maternal health information access.
- Fine-tuned large language models for Vietnamese medical question answering and adapted model responses to domain-specific healthcare content.

- Built a retrieval-augmented generation (RAG) pipeline to retrieve trusted medical knowledge, generate context-aware answers, and reduce hallucination in chatbot responses.
- Designed and implemented a simple web application for user interaction, chatbot demonstration, and project evaluation.

AI-Traffic-Analysis

- Project page: <https://github.com/tungedng2710/AI-Traffic-Analysis>
- Developed an end-to-end traffic video analysis system for vehicle detection, vehicle tracking, license plate detection, and license plate OCR.
- Trained and evaluated object detection and OCR models on traffic image datasets to improve recognition performance in real-world road scenes.
- Built a backend analysis service that integrates detection, tracking, plate localization, OCR, and result aggregation into a unified processing pipeline.
- Implemented a web-based demo application to visualize traffic analysis results and demonstrate the full computer vision workflow.

Vietnamese ID Card Recognition

- Researched and implemented object detection models for Vietnamese ID card localization and alignment in eKYC workflows.
- Developed preprocessing steps for document cropping, perspective correction, and image quality enhancement to improve downstream recognition accuracy.
- Integrated text detection and OCR models to extract identity information from Vietnamese ID card images.
- Evaluated recognition results and refined the processing pipeline for more robust performance under varied capture conditions.

CERTIFICATIONS

Quant Finance Explorer Program’24 , WorldQuant	11/2024
Google Advanced Data Analytics , Google	08/2024
Google Project Management , Google	08/2024
Google AI Essential , Google	08/2024
IELTS Academic , British Council, Overall score: 6.0	05/2023

ACTIVITIES

Mentoring Viettel Digital Talent Program	2026
Mentoring Scientific Research Competition for high school students	2023, 2024, 2025
Extra-class Assistant at the SAMI (HUST) study support club	01/2018 – 08/2018
Volunteer at the SAMI (HUST) Volunteer organization	10/2017 – 08/2018